1. Introduction

This report is focused on the network design for NetHub Company with seven branches located in Vienna, Morocco, Tunis, London, Paris, Rome, and Milan where there will be the data center and the headquarters.

2. Designing the network

The company has 7 branches, we need 8 LAN subnets and 7 WAN subnets. Each branch has only one subnet except Milan branch it will has 2 subnets, one for the Data Center, and one for the HQ. The connection between locations uses point-to-point topology each pair of routers are connected to each other, each router is connected to it's neighbour router. The connection inside each location uses a star topology except the Milan location, it uses extended star topology. Each device in the locations including pc's, laptops, and printers are connected to central switch.

2.1 Subnetting for LAN's

I made the subnetting for LAN's based on the business requirements, while I should take future expansion into account, each location employee numbers are not expected to be more than 500, so I made the subnetting of the private IP 10.0.0.0/8 based on the host number which is 500.

Subnetting:

- Specifying network and hosts bits:

  10.0.0.0/8

  N.H.H..H

- Using The Formula to specify number of bits needed for 500 hosts:
  2^x – 2 = Host Num
  2^x = 512
  x = 9 bits for the host

  10.0.0.0/23
  N.N.nnnn nnnh.hhhh hhhh

Because when 2^x – 2 = 500, x can't be a fixed number so I choose the number of host to be 512 so can x be a fixed number. So the total number of usable addresses for devices is 510 and two for the network id and broadcast ip.

Subnet	Branch	Network Id	First Ip	Last Ip	Broadcast Ip	Prefix
Sub 0	Milan DC	10.0.0.0	10.0.0.1	10.0.1.254	10.0.1.255	23
Sub 1	Milan HQ	10.0.2.0	10.0.2.1	10.0.3.254	10.0.3.255	23
Sub 2	Vienna	10.0.4.0	10.0.4.1	10.0.5.254	10.0.5.255	23
Sub 3	Morocco	10.0.6.0	10.0.6.1	10.0.7.254	10.0.7.255	23
Sub 4	Tunis	10.0.8.0	10.0.8.1	10.0.9.254	10.0.9.255	23
Sub 5	London	10.0.10.0	10.0.10.1	10.0.11.254	10.0.11.255	23
Sub 6	Paris	10.0.12.0	10.0.12.1	10.0.13.254	10.0.13.255	23
Sub 7	Rome	10.0.14.0	10.0.14.1	10.0.15.254	10.0.14.255	23

Subnet Mask: 255.255.254.0

Wild Card: 0.0.1.255

2.2 Subnetting For WAN’S

I made the subnetting for WAN’s based on the point-to-point topology between each pair of routers, so I made the subnetting of the 200.0.0.0/24

- Specifying network and hosts bits:
  200.0.0.0/24
  N.N.N.H
- Using The Formula to specify number of bits needed for 2hosts:
  2^x – 2 = Host Num
  2^x = 4
  x = 2 bits for the host

200.0.0.0/30

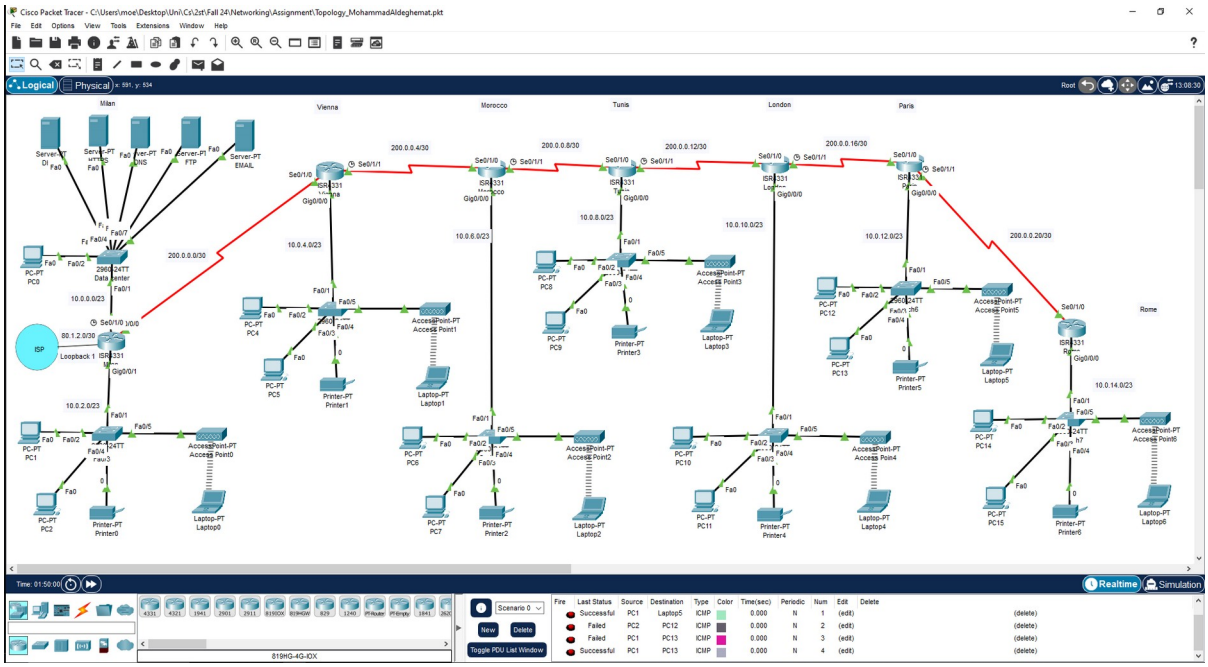
N.N.N.nnnn nnhh

Subnet	Link	Network Id	First Ip	Last Ip	Broadcast Ip	Prefix
Sub 0	Milan - Vienna	200.0.0.0	200.0.0.1	200.0.0.2	200.0.0.3	30
Sub 1	Vienna- Morocco	200.0.0.4	200.0.0.5	200.0.0.6	200.0.0.7	30
Sub 2	Morocco - Tunis	200.0.0.8	200.0.0.9	200.0.0.10	200.0.0.11	30
Sub 3	Tunis - London	200.0.0.12	200.0.0.13	200.0.0.14	200.0.0.15	30
Sub 4	London - Paris	200.0.0.16	200.0.0.17	200.0.0.18	200.0.0.19	30
Sub 5	Paris - Rome	200.0.0.20	200.0.0.21	200.0.0.22	200.0.0.23	30

Subnet Mask: 255.255.255.252

Wild Card: 0.0.0.3

3. Network Blueprint and Devices Configuration



- In Milan data center the first 10 Ip's are reserved for servers.
- In all remote offices the first 6 Ip's are reserved for devices like printers.
- All router's interface's were configured statically.

Milan:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
DHCP Server	10.0.0.2	255.255.254.0	23	Static
HTTPS Server	10.0.0.3	255.255.254.0	23	Static
DNS Server	10.0.0.4	255.255.254.0	23	Static
FTP Server	10.0.0.5	255.255.254.0	23	Static
EMAIL Server	10.0.0.6	255.255.254.0	23	Static
Pc0	10.0.0.12	255.255.254.0	23	Dynamic

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc1	10.0.2.8	255.255.254.0	23	Dynamic
Pc2	10.0.2.9	255.255.254.0	23	Dynamic
Laptop0	10.0.2.10	255.255.254.0	23	Dynamic
Printer0	10.0.2.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.0.1	255.255.254.0	23
Gig0/0/1	10.0.2.1	255.255.254.0	23
Se0/1/0	200.0.0.1	255.255.255.252	30

Vienna:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc4	10.0.4.8	255.255.254.0	23	Dynamic
Pc5	10.0.4.9	255.255.254.0	23	Dynamic
Laptop1	10.0.4.10	255.255.254.0	23	Dynamic
Printer1	10.0.4.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.4.1	255.255.254.0	23
Se0/1/0	200.0.0.2	255.255.255.252	30
Se0/1/1	200.0.0.5	255.255.255.252	30

Morocco:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc6	10.0.6.8	255.255.254.0	23	Dynamic
Pc7	10.0.6.9	255.255.254.0	23	Dynamic
Laptop2	10.0.6.10	255.255.254.0	23	Dynamic
Printer2	10.0.6.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.6.1	255.255.254.0	23
Se0/1/0	200.0.0.6	255.255.255.252	30
Se0/1/1	200.0.0.9	255.255.255.252	30

Tunis:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc8	10.0.8.8	255.255.254.0	23	Dynamic
Pc9	10.0.8.9	255.255.254.0	23	Dynamic
Laptop3	10.0.8.10	255.255.254.0	23	Dynamic
Printer3	10.0.8.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.8.1	255.255.254.0	23
Se0/1/0	200.0.0.10	255.255.255.252	30
Se0/1/1	200.0.0.13	255.255.255.252	30

London:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc10	10.0.10.8	255.255.254.0	23	Dynamic
Pc11	10.0.10.9	255.255.254.0	23	Dynamic
Laptop4	10.0.10.10	255.255.254.0	23	Dynamic
Printer4	10.0.10.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.10.1	255.255.254.0	23
Se0/1/0	200.0.0.14	255.255.255.252	30
Se0/1/1	200.0.0.17	255.255.255.252	30

Paris:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc12	10.0.12.8	255.255.254.0	23	Dynamic
Pc13	10.0.12.9	255.255.254.0	23	Dynamic
Laptop5	10.0.12.10	255.255.254.0	23	Dynamic
Printer5	10.0.12.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.12.1	255.255.254.0	23
Se0/1/0	200.0.0.18	255.255.255.252	30
Se0/1/1	200.0.0.21	255.255.255.252	30

Rome:

Device	Ip Address	Subnet Mask	Prefix	Dynamic/Static
Pc14	10.0.14.8	255.255.254.0	23	Dynamic
Pc15	10.0.14.9	255.255.254.0	23	Dynamic
Laptop6	10.0.14.10	255.255.254.0	23	Dynamic
Printer6	10.0.14.2	255.255.254.0	23	Static

Interface	Ip Address	Subnet Mask	Prefix
Gig0/0/0	10.0.14.1	255.255.254.0	23
Se0/1/0	200.0.0.22	255.255.255.252	30

For each router interface I used a static Ip configuration, and I enabled the OSPF (Open Shortest Path First) routing protocol to exchange routing information between routers and the OSPF has features such as:

- High scalability
- Fast Convergence
- More secure than RIP
- Higher bandwidth than RIP

I Secured each router by setting a password for:

- Console password: moe0000
- Privilege Mode Password: moeSec
- Privilege Mode Encrypted Password: moe2005

Router’s Configuration:

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0/0
Router(config-if)#ip address 10.0.0.1 255.255.254.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gig0/0/1
Router(config-if)#ip address 10.0.2.1 255.255.254.0
Router(config-if)#no shut
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/1/0
Router(config-if)#ip address 200.0.0.1 ?
  A.B.C.D  IP subnet mask
Router(config-if)#ip address 200.0.0.1 255.255.255.252
Router(config-if)#no-sh
Router(config-if)#no-shut
Router(config-if)#no shut
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#
```

```
Router(config)#interface loopback 1
Router(config-if)#ip address 80.1.2.1 255.255.255.252
Router(config-if)#no shutdow
Router(config-if)#no shutdown

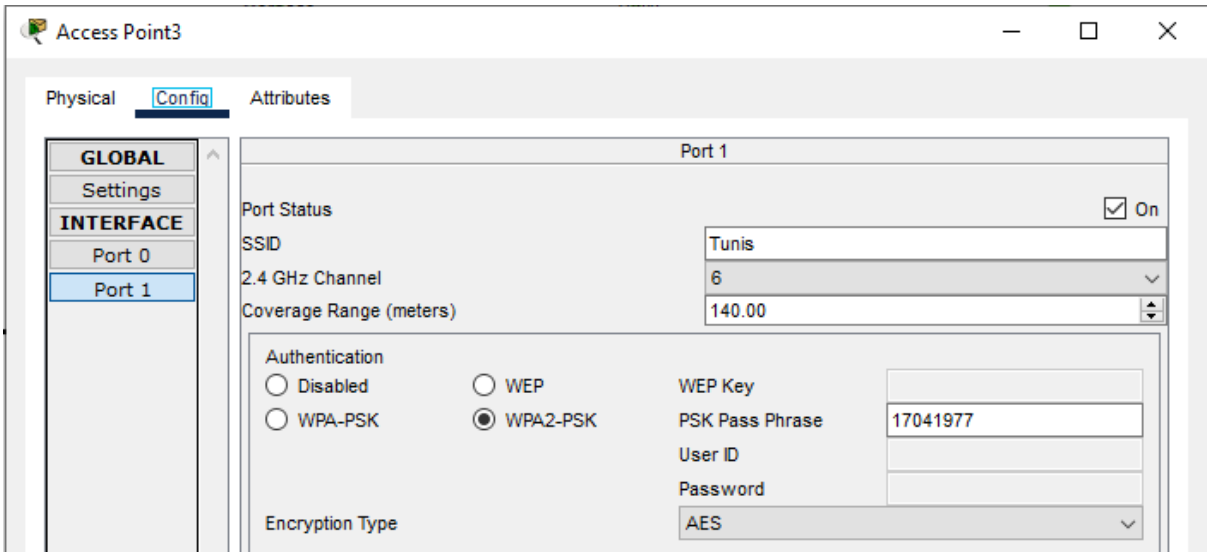
Router(config-if)#
%LINK-5-CHANGED: Interface Loopback1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
```

```
Router(config)#router ospf 100
Router(config-router)#pass
Router(config-router)#passive-interface gig0/0/0
Router(config-router)#passive-interface gig0/0/1
Router(config-router)#network 10.0.0.0 0.0.1.255 area 0
Router(config-router)#network 10.0.2.0 0.0.1.255 area 0
Router(config-router)#network 200.0.0.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#
```

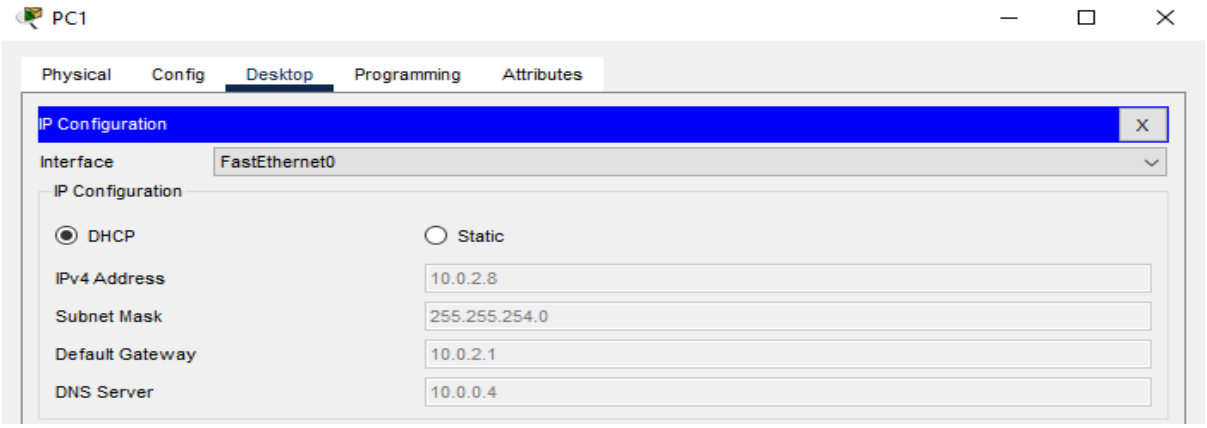
```
Router#
Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0/1
Router(config-if)#ip helper-
Router(config-if)#ip helper-address 10.0.0.2
Router(config-if)#
```

Access Point Configuration:

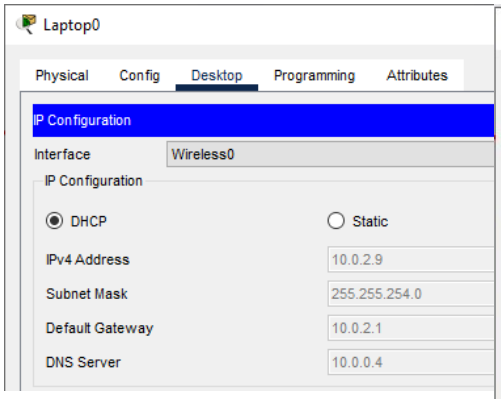


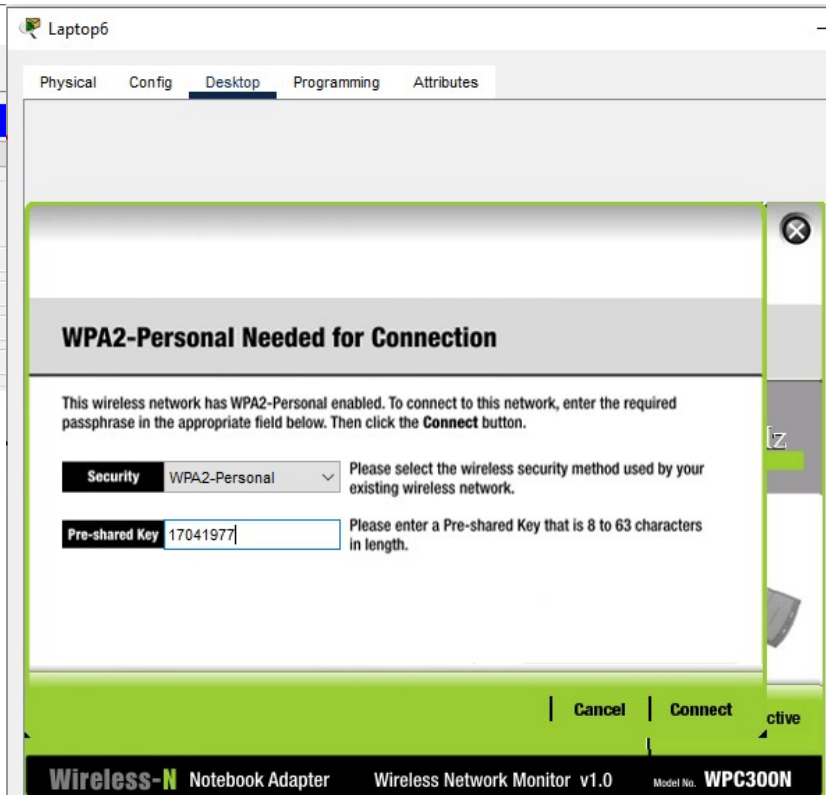
Access points password: 17041977

Pc's Configuration:

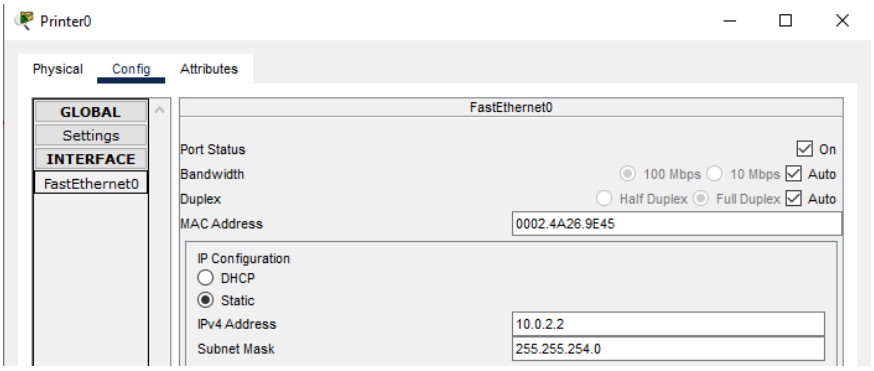


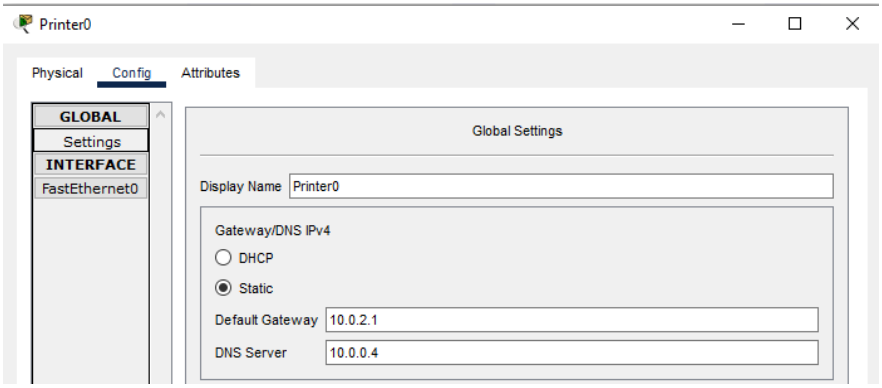
Laptop’s Configuration:





Printer’s Configuration:





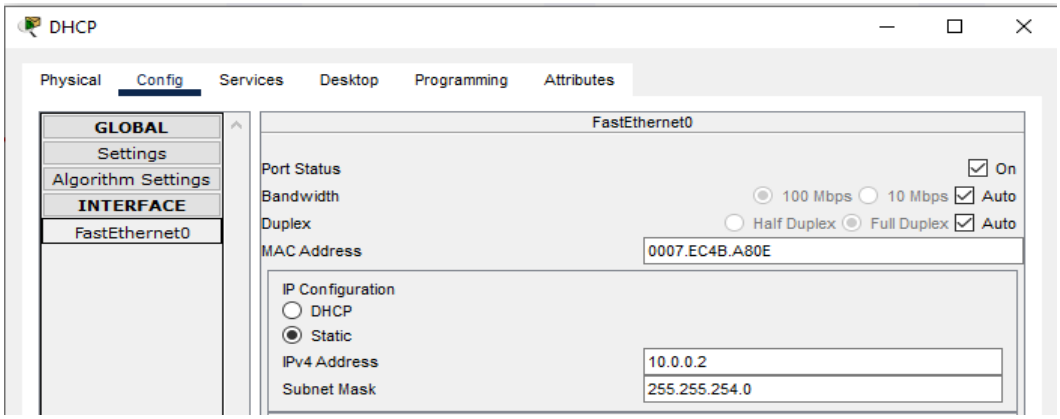
4. Server Installation and Services

In my design I installed 5 servers each with one service including HTTPS, DHCP, DNS, FTP, and Email. Each server has only one service running because if I installed the services on one server there is a single point of failure for services, if the server went down, all service will be down too. I used static Ip addressing for these servers and put the default gateway and DNS server. These servers are installed in the main data center in Milan branch.

- DHCP (Dynamic host configuration protocol): This service automatically assigns Ip addresses for devices, it is a crucial service that ensure that every device allocate a valid Ip address, it saves time and avoid Ip conflictions. It provides scalability and it is supported by most operating systems such as windows servers and Linux. The DHCP server have pools that are set manually which assigns Ip addresses from these pools.

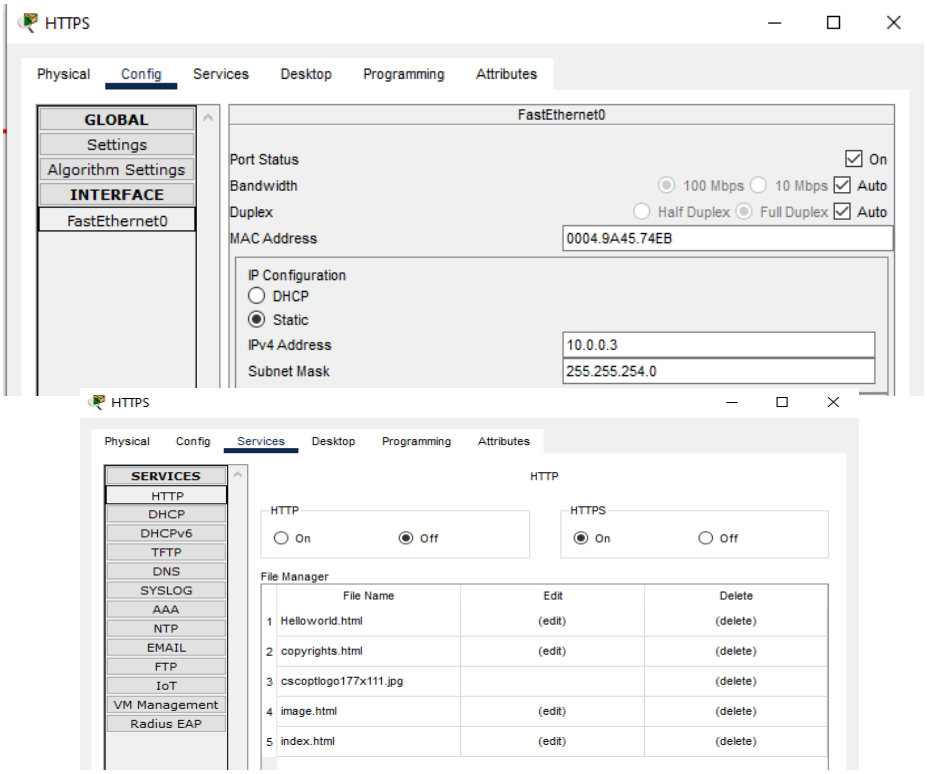
  In my design I installed DHCP service within its server, it has 7 pools, one pool for each LAN and it give Ip addresses based on these pools.

  Here is the configuration for DHCP server:

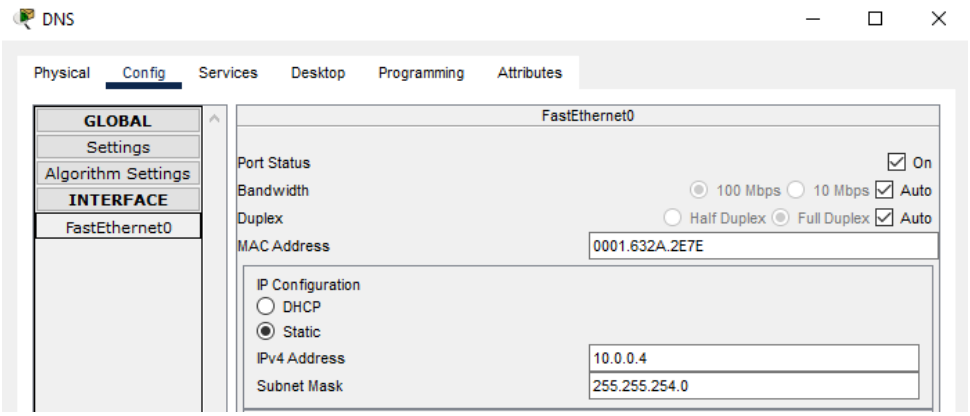


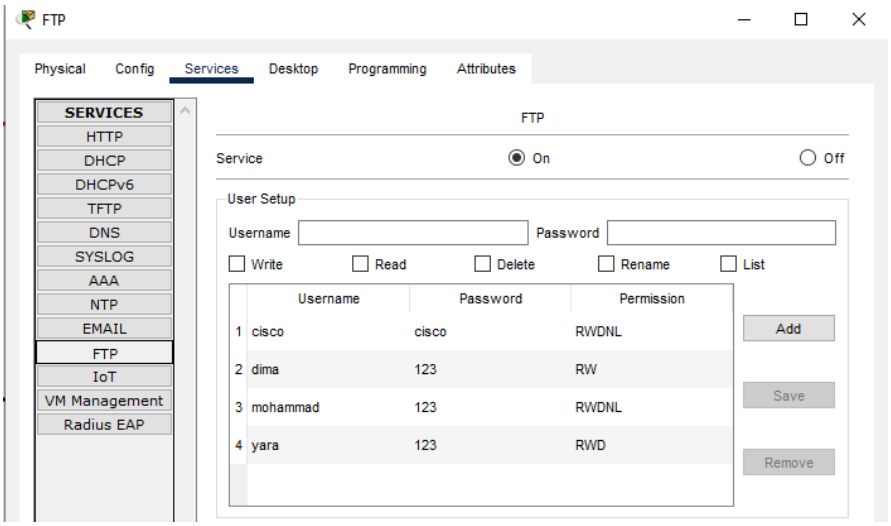
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
MilanDataCenter	10.0.0.1	10.0.0.4	10.0.0.12	255.255....	500	0.0.0.0	0.0.0.0
Rome	10.0.14.1	10.0.0.4	10.0.14.8	255.255....	500	0.0.0.0	0.0.0.0
Paris	10.0.12.1	10.0.0.4	10.0.12.8	255.255....	500	0.0.0.0	0.0.0.0
London	10.0.10.1	10.0.0.4	10.0.10.8	255.255....	500	0.0.0.0	0.0.0.0
Tunis	10.0.8.1	10.0.0.4	10.0.8.8	255.255....	500	0.0.0.0	0.0.0.0
Morocco	10.0.6.1	10.0.0.4	10.0.6.8	255.255....	500	0.0.0.0	0.0.0.0
Vienna	10.0.4.1	10.0.0.4	10.0.4.8	255.255....	500	0.0.0.0	0.0.0.0
MilanHQ	10.0.2.1	10.0.0.4	10.0.2.8	255.255....	500	0.0.0.0	0.0.0.0

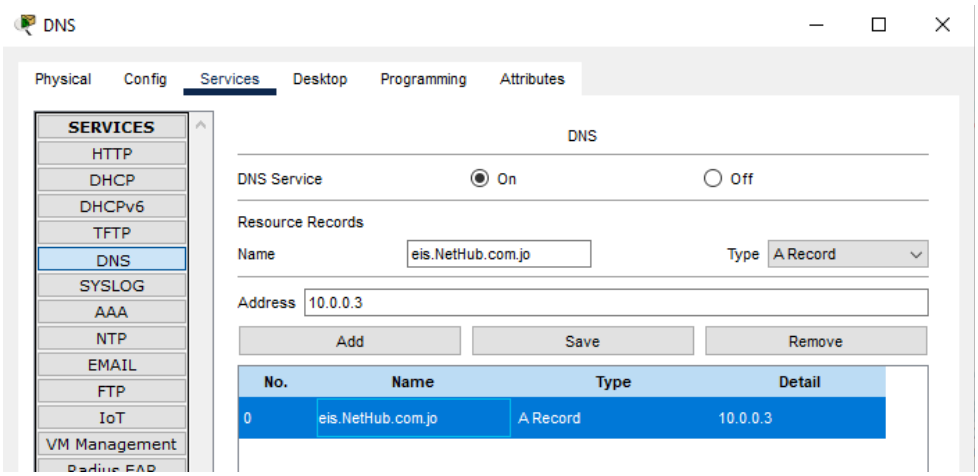
- HTTPS (Hyper Text Transfer Protocol Secure): A service used to communicate between devices and browser, this is a crucial service for the company to access its internal system. In my design, I installed HTTPS service on a dedicated server, and I choose HTTPS over HTTP because HTTPS provide more secure access to the company's system.
  HTTPS configuration:



- DNS (Domain Name Service): A service that do mapping between Ip address and the domain name, for example, it converts 10.0.0.3 to (eis.nethub.com.jo). Making it easier for employees to access the internal system.
  In my design, I installed DNS service within its server and here is the configuration for the DNS server:





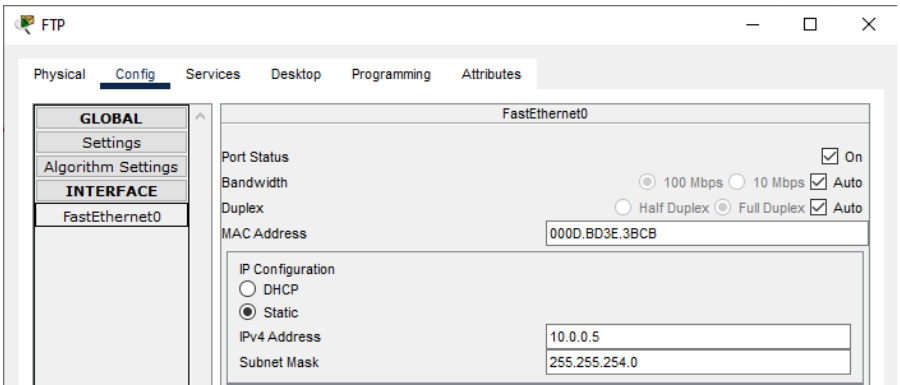


- FTP (File Transfer Protocol): A service that provide the means to transfer files between devices in the network, employees can upload (push) and download (pull) files.

  In my design I installed FTP service within its server, I made three users (mohammad, yara, dima) with a password (123) and here is the configuration for FTP server:

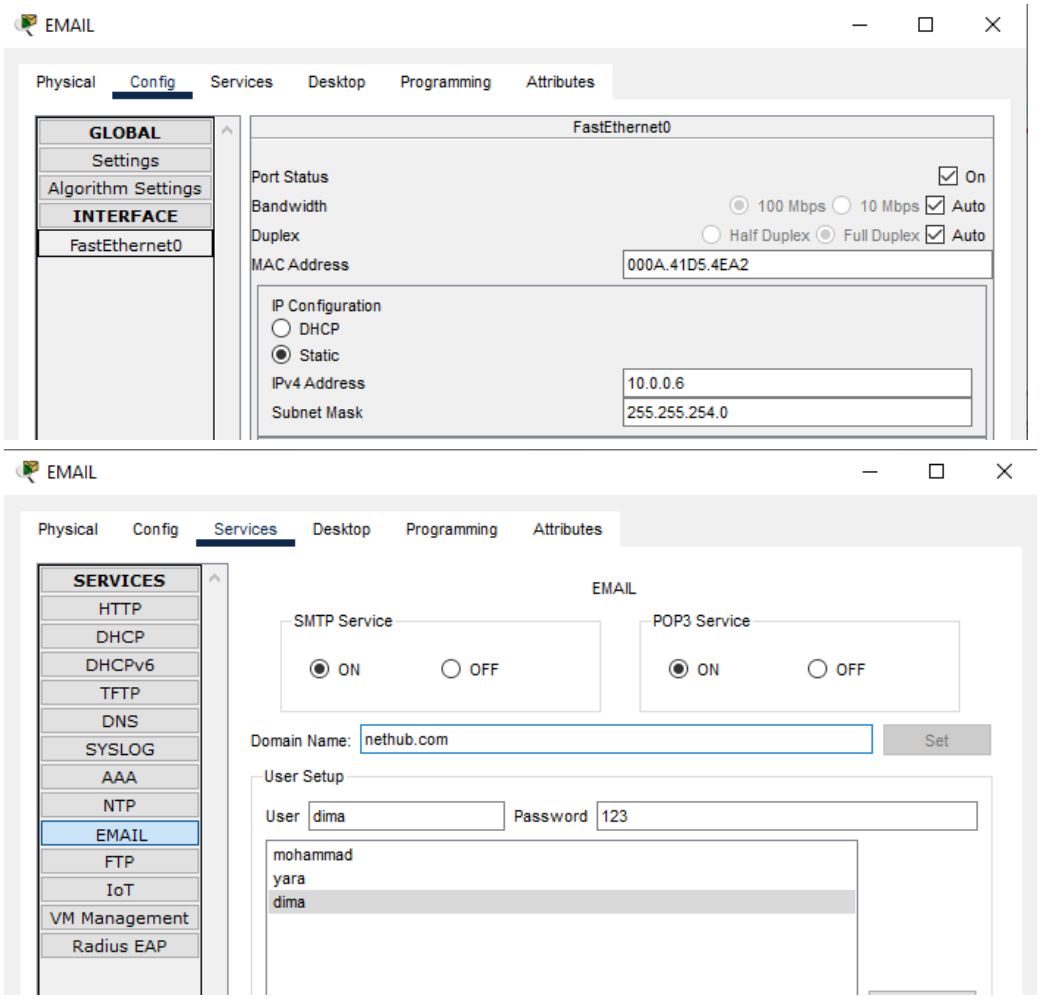
- EMAIL: A service that is responsible for the email sending and receiving between devices in the network. Email service includes:

  1. SMTP (Simple Mail Transfer Protocol): It is for sending emails.



2. POP3 (Post Office Protocol Version 3): It is for retrieving emails from email server.

In my design I installed Email service within its server, I made three users (moe, yara, dima) with a password (123) and here is the configuration for Email server:

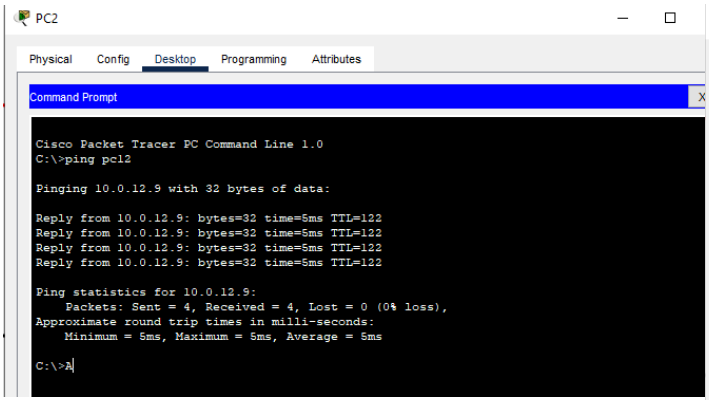


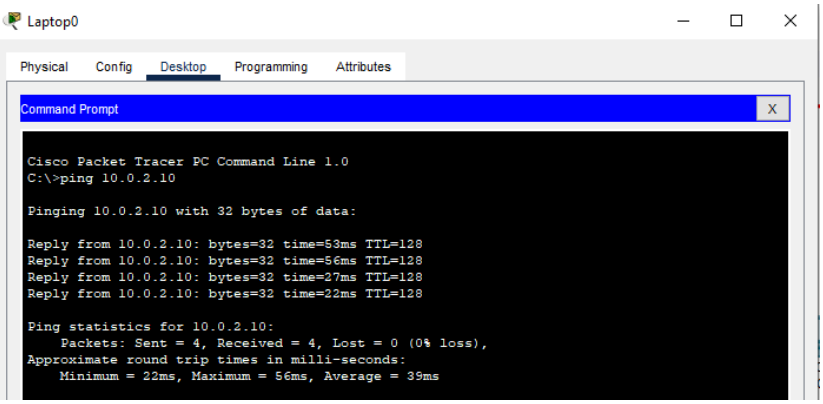
5. Test Results

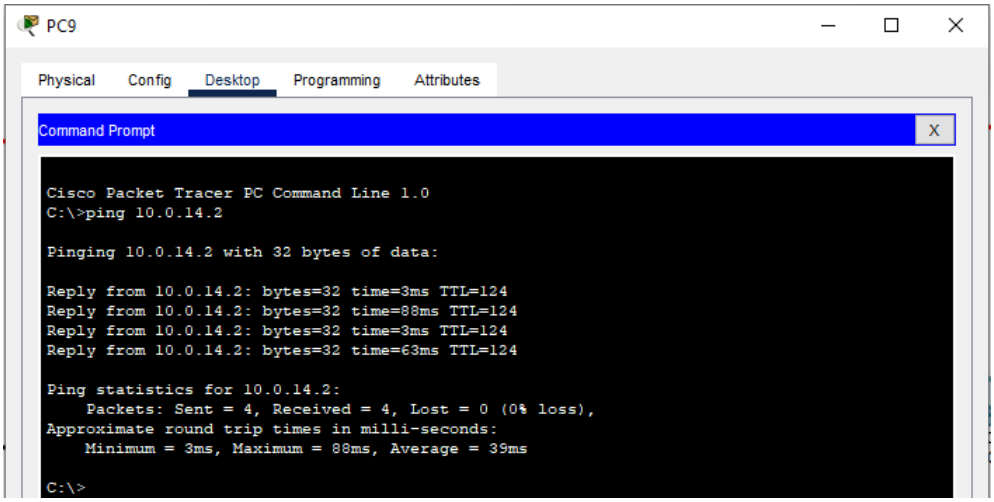
I did the verification and test based on the test plan, and I focused on the connectivity, service availability, routing protocol, performance.

- Connectivity: In this part we will test connectivity between devices in the same network and devices in a remote network. The connectivity in my design appears that it works accurately and doesn't have any error

  The first test is a test using ping command from pc2 in Milan to pc12 in Paris.
  The second test is a test using ping command from laptop0 in Milan to pc1 in Milan
  The third test is a test using ping command from pc9 in Tunis to printer6 in Rome.


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping pc12
Pinging 10.0.12.9 with 32 bytes of data:
Reply from 10.0.12.9: bytes=32 time=5ms TTL=122
Reply from 10.0.12.9: bytes=32 time=5ms TTL=122
Reply from 10.0.12.9: bytes=32 time=5ms TTL=122
Reply from 10.0.12.9: bytes=32 time=5ms TTL=122
Ping statistics for 10.0.12.9:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 5ms, Average = 5ms
C:\>


Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.2.10
Pinging 10.0.2.10 with 32 bytes of data:
Reply from 10.0.2.10: bytes=32 time=53ms TTL=128
Reply from 10.0.2.10: bytes=32 time=56ms TTL=128
Reply from 10.0.2.10: bytes=32 time=27ms TTL=128
Reply from 10.0.2.10: bytes=32 time=22ms TTL=128
Ping statistics for 10.0.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 22ms, Maximum = 56ms, Average = 39ms


PC9
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.14.2
Pinging 10.0.14.2 with 32 bytes of data:
Reply from 10.0.14.2: bytes=32 time=3ms TTL=124
Reply from 10.0.14.2: bytes=32 time=88ms TTL=124
Reply from 10.0.14.2: bytes=32 time=3ms TTL=124
Reply from 10.0.14.2: bytes=32 time=63ms TTL=124
Ping statistics for 10.0.14.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 88ms, Average = 39ms
C:\>

- Routers: to test that routers work correctly first we will see the routing table by the command “show Ip route”

Milan

Physical Config CLI Attributes

IOS Command Line Interface

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.0.0.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
C       10.0.2.0/23 is directly connected, GigabitEthernet0/0/1
L       10.0.2.1/32 is directly connected, GigabitEthernet0/0/1
O       10.0.4.0/23 [110/65] via 200.0.0.2, 00:23:52, Serial0/1/0
O       10.0.6.0/23 [110/129] via 200.0.0.2, 00:23:52, Serial0/1/0
O       10.0.8.0/23 [110/193] via 200.0.0.2, 00:23:52, Serial0/1/0
O       10.0.10.0/23 [110/257] via 200.0.0.2, 00:23:52, Serial0/1/0
O       10.0.12.0/23 [110/321] via 200.0.0.2, 00:23:52, Serial0/1/0
O       10.0.14.0/23 [110/385] via 200.0.0.2, 00:23:52, Serial0/1/0
     80.0.0.0/8 is variably subnetted, 2 masks
C       80.1.2.0/30 is directly connected, Loopback1
L       80.1.2.1/32 is directly connected, Loopback1
     200.0.0.0/24 is variably subnetted, 7 subnets, 2 masks
C       200.0.0.0/30 is directly connected, Serial0/1/0
L       200.0.0.1/32 is directly connected, Serial0/1/0
O       200.0.0.4/30 [110/128] via 200.0.0.2, 00:23:52, Serial0/1/0
O       200.0.0.8/30 [110/192] via 200.0.0.2, 00:23:52, Serial0/1/0
O       200.0.0.12/30 [110/256] via 200.0.0.2, 00:23:52, Serial0/1/0
O       200.0.0.16/30 [110/320] via 200.0.0.2, 00:23:52, Serial0/1/0
O       200.0.0.20/30 [110/384] via 200.0.0.2, 00:23:52, Serial0/1/0
S*   0.0.0.0/0 is directly connected, Loopback1
```

Vienna

Physical Config CLI Attributes

IOS Command Line Interface

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.0.0.0/23 [110/65] via 200.0.0.1, 00:26:48, Serial0/1/0
O       10.0.2.0/23 [110/65] via 200.0.0.1, 00:26:48, Serial0/1/0
C       10.0.4.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.4.1/32 is directly connected, GigabitEthernet0/0/0
O       10.0.6.0/23 [110/65] via 200.0.0.6, 00:26:48, Serial0/1/1
O       10.0.8.0/23 [110/129] via 200.0.0.6, 00:26:38, Serial0/1/1
O       10.0.10.0/23 [110/193] via 200.0.0.6, 00:26:38, Serial0/1/1
O       10.0.12.0/23 [110/257] via 200.0.0.6, 00:26:38, Serial0/1/1
O       10.0.14.0/23 [110/321] via 200.0.0.6, 00:26:38, Serial0/1/1
     200.0.0.0/24 is variably subnetted, 8 subnets, 2 masks
C       200.0.0.0/30 is directly connected, Serial0/1/0
L       200.0.0.2/32 is directly connected, Serial0/1/0
C       200.0.0.4/30 is directly connected, Serial0/1/1
L       200.0.0.5/32 is directly connected, Serial0/1/1
O       200.0.0.8/30 [110/128] via 200.0.0.6, 00:26:48, Serial0/1/1
O       200.0.0.12/30 [110/192] via 200.0.0.6, 00:26:38, Serial0/1/1
O       200.0.0.16/30 [110/256] via 200.0.0.6, 00:26:38, Serial0/1/1
O       200.0.0.20/30 [110/320] via 200.0.0.6, 00:26:38, Serial0/1/1
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

Morocco

Physical Config CLI Attributes

IOS Command Line Interface

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.0.0.0/23 [110/129] via 200.0.0.5, 00:32:05, Serial0/1/0
O       10.0.2.0/23 [110/129] via 200.0.0.5, 00:32:05, Serial0/1/0
O       10.0.4.0/23 [110/65] via 200.0.0.5, 00:32:15, Serial0/1/0
C       10.0.6.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.6.1/32 is directly connected, GigabitEthernet0/0/0
O       10.0.8.0/23 [110/65] via 200.0.0.10, 00:32:15, Serial0/1/1
O       10.0.10.0/23 [110/129] via 200.0.0.10, 00:32:05, Serial0/1/1
O       10.0.12.0/23 [110/193] via 200.0.0.10, 00:32:05, Serial0/1/1
O       10.0.14.0/23 [110/257] via 200.0.0.10, 00:32:05, Serial0/1/1
     200.0.0.0/24 is variably subnetted, 8 subnets, 2 masks
O       200.0.0.0/30 [110/128] via 200.0.0.5, 00:32:15, Serial0/1/0
C       200.0.0.4/30 is directly connected, Serial0/1/0
L       200.0.0.6/32 is directly connected, Serial0/1/0
C       200.0.0.8/30 is directly connected, Serial0/1/1
L       200.0.0.9/32 is directly connected, Serial0/1/1
O       200.0.0.12/30 [110/128] via 200.0.0.10, 00:32:15, Serial0/1/1
O       200.0.0.16/30 [110/192] via 200.0.0.10, 00:32:05, Serial0/1/1
O       200.0.0.20/30 [110/256] via 200.0.0.10, 00:32:05, Serial0/1/1
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

Tunis

Physical Config CLI Attributes

IOS Command Line Interface

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.0.0.0/23 [110/193] via 200.0.0.9, 00:32:45, Serial0/1/0
O       10.0.2.0/23 [110/193] via 200.0.0.9, 00:32:45, Serial0/1/0
O       10.0.4.0/23 [110/129] via 200.0.0.9, 00:32:45, Serial0/1/0
O       10.0.6.0/23 [110/65] via 200.0.0.9, 00:32:45, Serial0/1/0
C       10.0.8.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.8.1/32 is directly connected, GigabitEthernet0/0/0
O       10.0.10.0/23 [110/65] via 200.0.0.14, 00:32:45, Serial0/1/1
O       10.0.12.0/23 [110/129] via 200.0.0.14, 00:32:45, Serial0/1/1
O       10.0.14.0/23 [110/193] via 200.0.0.14, 00:32:45, Serial0/1/1
     200.0.0.0/24 is variably subnetted, 8 subnets, 2 masks
O       200.0.0.0/30 [110/192] via 200.0.0.9, 00:32:45, Serial0/1/0
O       200.0.0.4/30 [110/128] via 200.0.0.9, 00:32:45, Serial0/1/0
C       200.0.0.8/30 is directly connected, Serial0/1/0
L       200.0.0.10/32 is directly connected, Serial0/1/0
C       200.0.0.12/30 is directly connected, Serial0/1/1
L       200.0.0.13/32 is directly connected, Serial0/1/1
O       200.0.0.16/30 [110/128] via 200.0.0.14, 00:32:45, Serial0/1/1
O       200.0.0.20/30 [110/192] via 200.0.0.14, 00:32:45, Serial0/1/1
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

London

Physical Config CLI Attributes

IOS Command Line Interface

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.0.0.0/23 [110/257] via 200.0.0.13, 00:34:26, Serial0/1/0
O       10.0.2.0/23 [110/257] via 200.0.0.13, 00:34:26, Serial0/1/0
O       10.0.4.0/23 [110/193] via 200.0.0.13, 00:34:26, Serial0/1/0
O       10.0.6.0/23 [110/129] via 200.0.0.13, 00:34:26, Serial0/1/0
O       10.0.8.0/23 [110/65] via 200.0.0.13, 00:34:26, Serial0/1/0
C       10.0.10.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.10.1/32 is directly connected, GigabitEthernet0/0/0
O       10.0.12.0/23 [110/65] via 200.0.0.18, 00:34:26, Serial0/1/1
O       10.0.14.0/23 [110/129] via 200.0.0.18, 00:34:26, Serial0/1/1
     200.0.0.0/24 is variably subnetted, 8 subnets, 2 masks
O       200.0.0.0/30 [110/256] via 200.0.0.13, 00:34:26, Serial0/1/0
O       200.0.0.4/30 [110/192] via 200.0.0.13, 00:34:26, Serial0/1/0
O       200.0.0.8/30 [110/128] via 200.0.0.13, 00:34:26, Serial0/1/0
C       200.0.0.12/30 is directly connected, Serial0/1/0
L       200.0.0.14/32 is directly connected, Serial0/1/0
C       200.0.0.16/30 is directly connected, Serial0/1/1
L       200.0.0.17/32 is directly connected, Serial0/1/1
O       200.0.0.20/30 [110/128] via 200.0.0.18, 00:34:26, Serial0/1/1
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

Rome

Physical Config CLI Attributes

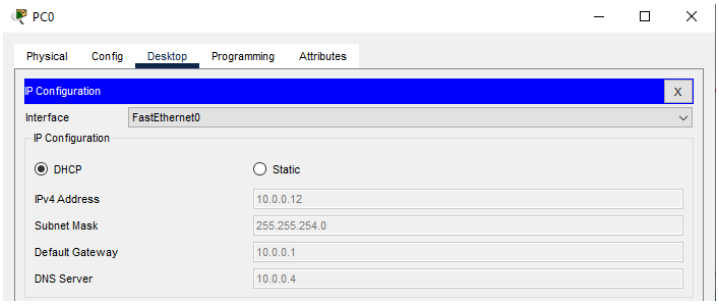
IOS Command Line Interface

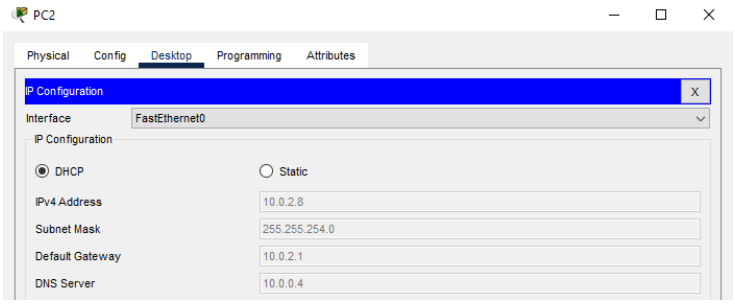
```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0

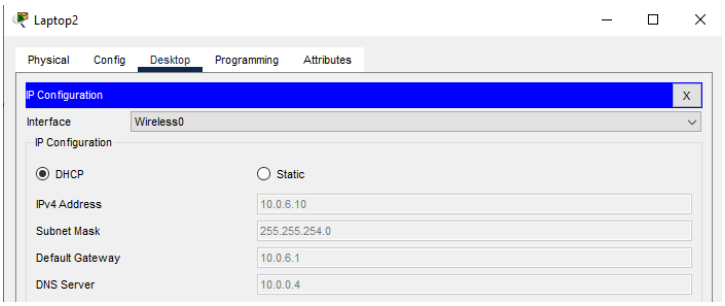
     10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       10.0.0.0/23 [110/385] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.2.0/23 [110/385] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.4.0/23 [110/321] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.6.0/23 [110/257] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.8.0/23 [110/193] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.10.0/23 [110/129] via 200.0.0.21, 00:34:39, Serial0/1/0
O       10.0.12.0/23 [110/65] via 200.0.0.21, 00:34:49, Serial0/1/0
C       10.0.14.0/23 is directly connected, GigabitEthernet0/0/0
L       10.0.14.1/32 is directly connected, GigabitEthernet0/0/0
     200.0.0.0/24 is variably subnetted, 7 subnets, 2 masks
O       200.0.0.0/30 [110/384] via 200.0.0.21, 00:34:39, Serial0/1/0
O       200.0.0.4/30 [110/320] via 200.0.0.21, 00:34:39, Serial0/1/0
O       200.0.0.8/30 [110/256] via 200.0.0.21, 00:34:39, Serial0/1/0
O       200.0.0.12/30 [110/192] via 200.0.0.21, 00:34:39, Serial0/1/0
O       200.0.0.16/30 [110/128] via 200.0.0.21, 00:34:49, Serial0/1/0
C       200.0.0.20/30 is directly connected, Serial0/1/0
L       200.0.0.22/32 is directly connected, Serial0/1/0
S*   0.0.0.0/0 is directly connected, Serial0/1/0
```

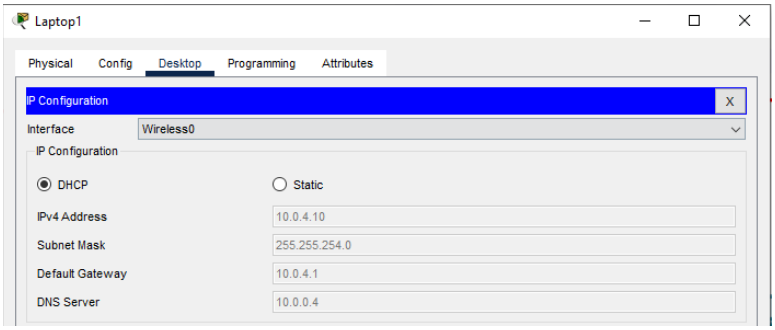
- Service Availability: we will test service availability in different ways depending on the service as we can see below.

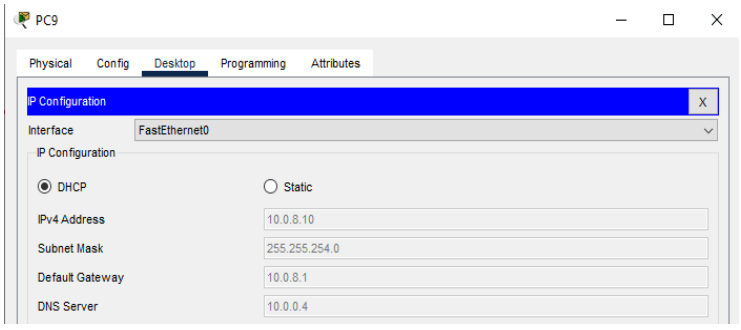
  DHCP: we will test service availability by entering one device in every remote network and give it a dynamic address.

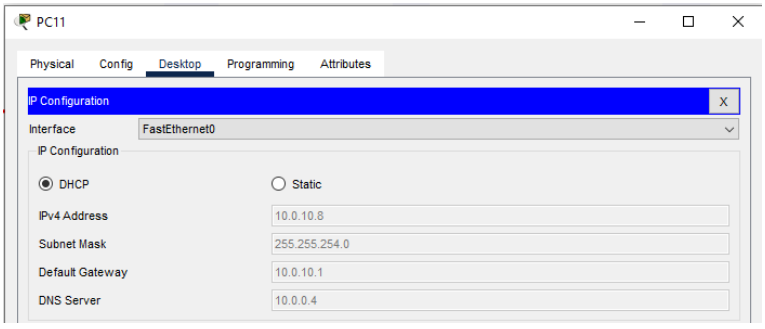


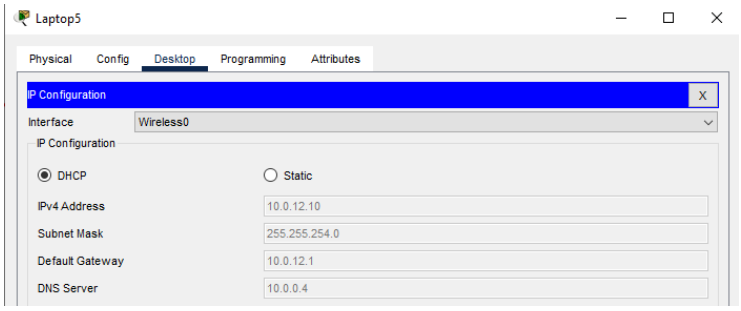


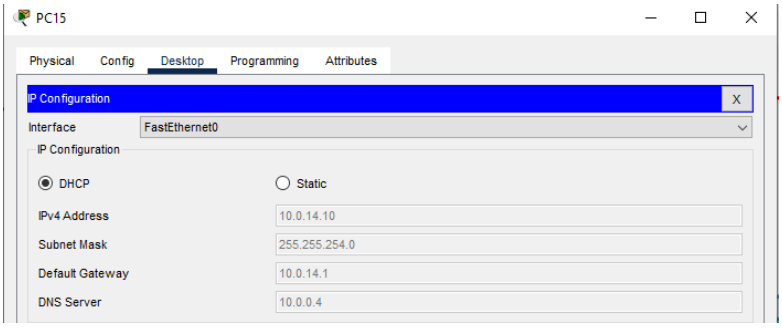




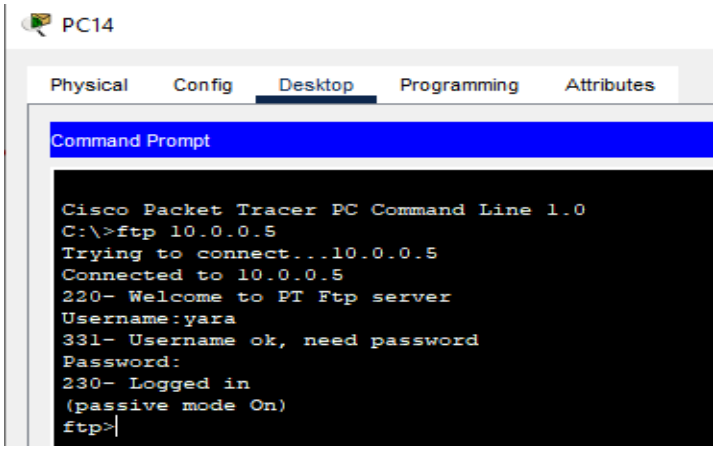




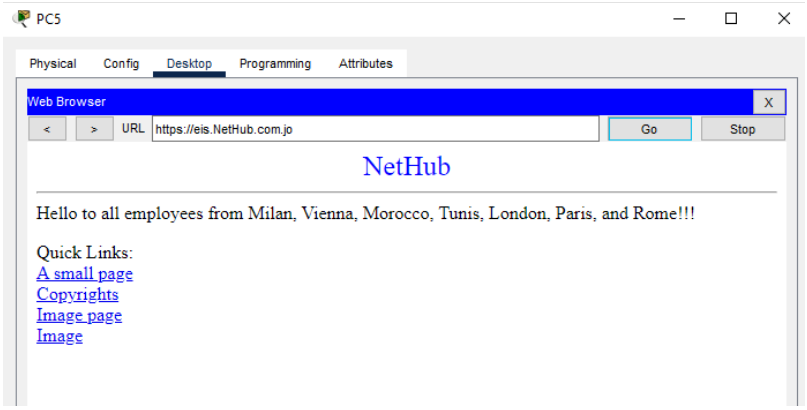


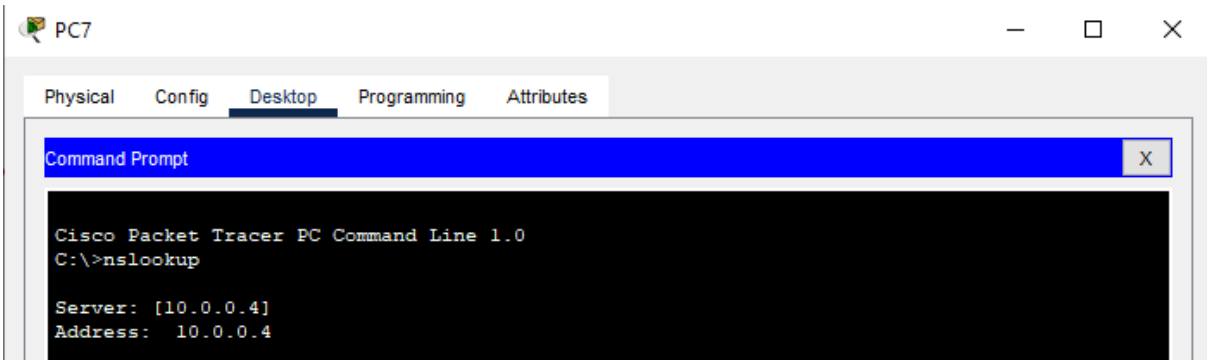


FTP: we will test service availability by connecting to ftp server.

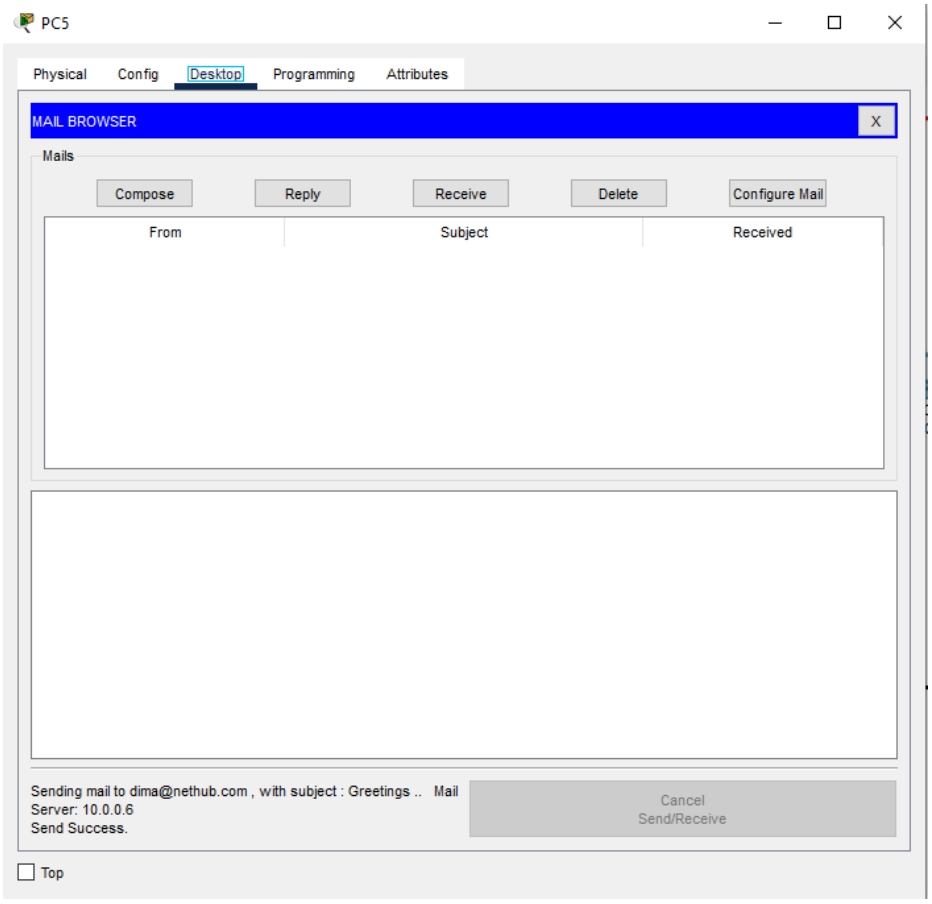


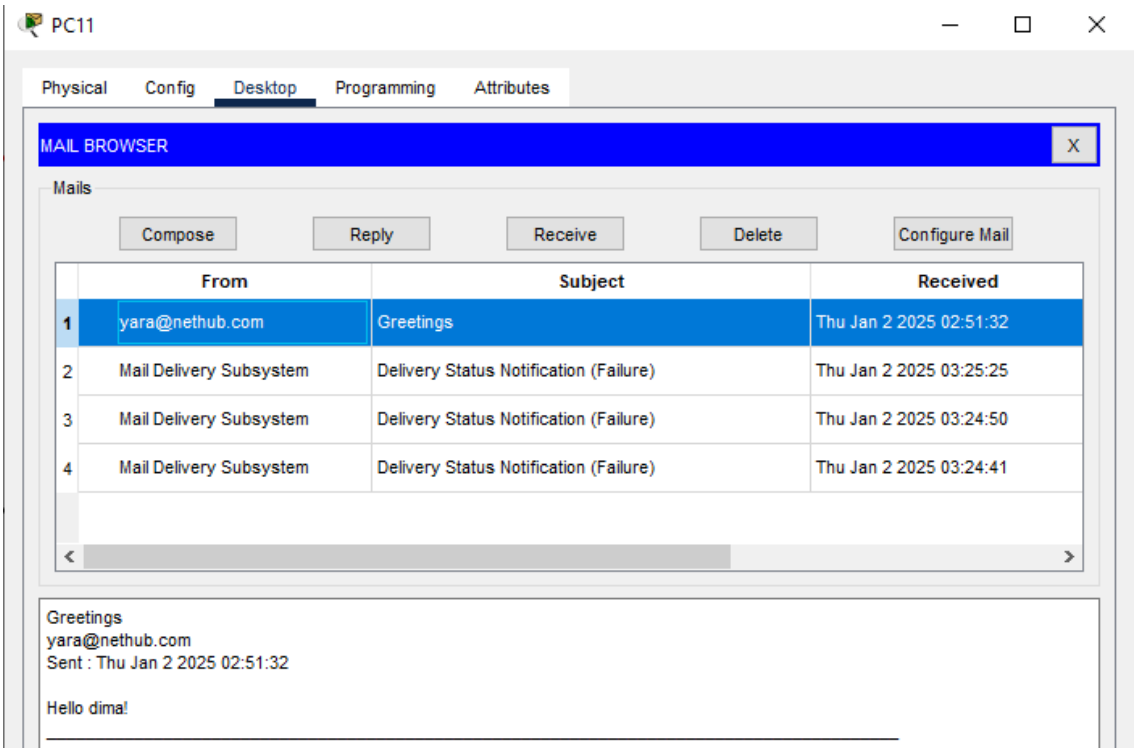
HTTPS and DNS: we will test browsing service by opening the web browser and type the IP or the domain name to test the DNS. Also we will test DNS service by the command “nslookup”.





Email: we will test the service availability by sending and receiving email from device to device. The failures were for test purposes.


PC5
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
Sending mail to dima@nethub.com , with subject : Greetings .. Mail Server: 10.0.0.6
Send Success.
Cancel
Send/Receive
Top


PC11
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1 yara@nethub.com Greetings Thu Jan 2 2025 02:51:32
2 Mail Delivery Subsystem Delivery Status Notification (Failure) Thu Jan 2 2025 03:25:25
3 Mail Delivery Subsystem Delivery Status Notification (Failure) Thu Jan 2 2025 03:24:50
4 Mail Delivery Subsystem Delivery Status Notification (Failure) Thu Jan 2 2025 03:24:41
Greetings
yara@nethub.com
Sent : Thu Jan 2 2025 02:51:32
Hello dima!

6. Critical Reflection

In my design I focused on making an efficient, and scalable network, as I made the subnets for future expansion as it has 500 employees in every network and there is 500 valid Ip address in the network, also I avoided waste of Ip addresses. I used a point-to-point for WAN and star topology for LAN as they help me to maintain an easy scalable network and they are easy to maintain, easy to troubleshoot, and cost effective.

Also, I made each service including DHCP, HTTPS, DNS, EMAIL, FTP installed on a dedicated server, means that DHCP on its own server, Email on its own server, and the same for other services. It was because I don't want all services to be in one server because it that server fails, all the services will fail.

I gave the servers and printers a static Ip address unlike other devices that were taken a dynamic Ip address ensuring that these devices will always have the same Ip address because if we give them a dynamic Ip address the Ip may change and then we change the Ip for these services in all devices.

The test plan was perfect to analyse what issues might been in the network, it helps me to make test to the network to see if everything works accurately.

The future plans were considered perfectly as the company requirements didn't want security, but this is a big challenge that should be considered in the future to make the network more secure because if the network is not secure as much it can be accessible by hackers and cyber-attacks.

The media used also should be considered to change them to Fiber-optic because it has more advantages that the media used.

Also, for the DHCP server I made the first 10 Ip addresses reserved for future servers because I will give every server a static Ip address, and the first 6 Ip addresses were reserved for devices like printers because I also will give devices like printers a static Ip address.

So, if the company expands, the network can handle more remote offices and more devices in these remote offices and new offices can easily be added without having to design a new network.

Overall, the design and implementation of the network were as good as possible, by addressing business requirements I was successfully able to design and implement a good network that supports business needs, business growth, and supports technological growth needs.